Journal of International Education and Development
2024, VOL. 8, NO. 5, 166-179
DOI: 10.47297/wspiedWSP2516-250029.20240805

Research on the Construction of School Enterprise Joint Construction Training Bases under the Background of Industry Education Integration

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ABSTRACT

This paper explores the significance of industry-university-research (IUR) cooperation base construction in the context of China's socio-economic development. It analyzes the necessity, key points, development directions, and experiences of IUR cooperation base construction both domestically and internationally, and finally proposes future research directions and prospects. The key Findings are as follows, necessity: IUR cooperation bases address issues in urban-rural development, industrial transformation, and talent cultivation, benefiting enterprises, universities, and governments; development directions: Focus on industry-education integration, local characteristics, and innovation-driven talent training; domestic and international models: Analyzes various IUR cooperation base models in China and abroad, highlighting successful practices and lessons learned; factors Influencing cooperation: Identifies key factors such as policy environment, cooperation mechanisms, economic support, and the strength of schools and enterprises; future research Directions: recommends establishing an IUR evaluation mechanism, improving cooperation base models, promoting technology transfer, and serving national development strategies.

KEYWORDS

Training Base; Industry-university-research; Practical experience

1 Introduction

1.1 Research Background

At present, the problem of urban and rural construction is an important problem in China's development, and the employment development during the transformation period is also an important problem. Financial support, policy platform, enterprise innovation and university employment are the important means to solve the bottleneck of urban and rural development. Cultivating innovative talents has been raised to the height of national strategy, and it is also one of the important means to improve China's comprehensive national strength[1]. The development is based on the concept of integrated development, to use vocational education to jointly create a new model of metropolitan vocational education integration and diversified collaborative education, which has a positive role.

At present, urban and rural construction has been in the transition stage, and the development

of digital technology has given a new development impetus to urban and rural construction. For the country, it is possible to actively build a practical platform integrating domestic enterprises, universities and research institutes and provide long-term and stable incentives to promote structural transformation. Through the cooperation between industry, university and research institutes, it can take the sustained and rapid development of economic subjects as the goal, effectively integrate social advantageous resources, improve the efficiency of scientific and technological input and technological innovation, promote the transformation of scientific and technological achievements into real productive forces, and make the economic subjects develop better and faster[2].

For enterprises, finding a sustainable business growth model is an important task for the development of enterprises. For students, the employment rate is relatively stable, but in fact, the real employment situation of graduates in universities has become difficult, industrial transformation and industry changes also have a great impact on students. For the overall development of urban and rural areas, China's urbanization rate has been close to 60%. With the deepening of the urbanization process, it is particularly necessary to promote rural revitalization and urban-rural integration.

Although the IUR education teaching has been explored to some extent, it is still lacking based on the digital practice. Urban and rural construction is not separated from civil engineering, architecture, geology, urban and rural planning and other students. With the combination of industry, university and research and urban and rural construction, the establishment of a think tank platform to bring students into field investigation, subject practice and subject teaching combination, and the scientific research and teaching base jointly can be built by university-enterprise cooperation. Finally, can the reliable effect and results of urban and rural construction be formed.

1.2 Purpose of research

It is of obvious theoretical and practical significance to promote the research on the construction of school-enterprise joint training base under the background of industry-education integration, Therefore, based on the specific practical experience research in colleges and universities, it can be realized that:

(1) Promote the in-depth development of the integration theory of industry and education. Through in-depth study on the construction of the training base, explore the best practice, cooperation mode and management mechanism of the integration of industry and education.

(2) Through the research on the construction of the practical training base under the background of industry-education integration, the existing work method can be verified and improved. This will help to improve the operability and applicability of the theory, and provide a more reliable theoretical support for the practical operation and policy making.

(3) Study the development path, policy and management coordination mechanism of urban and rural construction, as well as the relevant base construction conditions, requirements and implementation approaches, and help to provide empirical research results and guidance, and provide decision support and reference for government departments in the formulation of relevant policies and management practices.

(4) Research on the construction of university-enterprise joint training base plays an important role in promoting IUR cooperation.

Through in-depth research on the problems and challenges in practice, we will build a closer cooperation bridge between universities and enterprises, and promote knowledge transfer and technological innovation.

1.3 Research method

Literature analysis. Based on relevant tools such as CITESPACE and origin, the cutting-edge research results and literature are collected in core databases such as CNKI and Web of science, and the main direction and content of relevant research are clarified, and analyzed and summarized.

Example analysis method. Based on the schools of the teachers of the research team as an example, a comprehensive analysis summarizes the construction characteristics of relevant bases, and starting from the practical experience, the conditions, measures and specific contents of the construction of practice bases for school-enterprise cooperation should be concluded.

2 Key points and development directions of industry-university-research cooperation

At present, whether vocational education undergraduate, full-time undergraduate, as well as research education are emphasizing the integration of practice and practice. From the fund literature in CNKI, with "industry, university and research" and "practice base", a total of 430 existing articles. According to the content of the 430 documents, keywords and cooperative institutions were analyzed, among which 429 effective documents were analyzed.



Figure 1 Word frequency cloud map

First, a systematic analysis was conducted, making word frequency co-occurrence based on all keywords, and the keyword co-occurrence map in Figure 1 was obtained. It can be seen that the keywords "Practical teaching", "Personnel training", "university-industry cooperation", "IUR cooperation" and "practical teaching" appear frequently around the core word of "industry-university-research". It is a high-frequency vocabulary in the research process of major research scholars in recent years and represents a series of research directions. In practice, scientific consideration of new engineering, professional degree education and other contents has a profound impact on the construction of industry-university-research base, and has certain practical significance. However, in order to analyze the key points of the construction of industry-university-research base in detail, the main direction of the future construction of industry-university-research base, a deeper analysis of the connection and influence of key words was made. Therefore, the in-depth analysis based on Citespace can get results and get the keyword map in Figure 2.

2.1 Must based on the integration of industry and education, and promote the team building of double-teacher team and enterprise mentors

In order to promote the construction of industry-university and research bases and ensure that will achieves its effect, the fundamental place of the construction of industry-university and

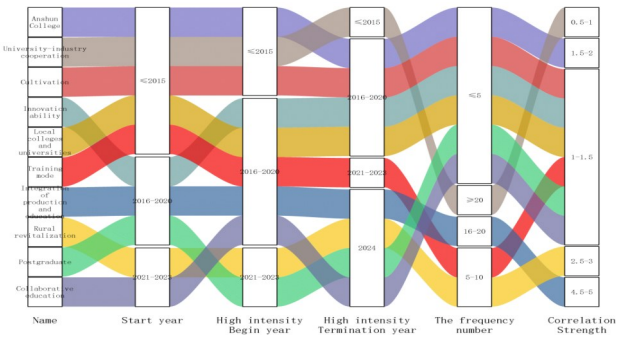


Figure 2 An alluvial map formed based on the reference analysis

research bases should be clearly defined. The essence industry-university cooperation is an effective combination of various production factors needed to promote technological innovation. Its core has three: first, to ensure that the development and construction of the industry can keep up with the pace of The Times, Ensure the competitiveness of the industry, at the same time, For enterprises to provide relevant talents that can match the positions, Reduce the energy and cost of enterprises in talent training; Second, to promote the combination of education and teaching curriculum system and industry development, While promoting the development of the industry, Optimize the system content of the course, Ensure the update of the knowledge system; Third, promote the development and innovation of university technology, Pilot new technologies and new theories in enterprise construction and development, To provide a guarantee for the implementation of university research, At the same time to ensure the knowledge, ability and technical ability of college students, Do not have the ability to post after graduation.

The core of the construction of IUR base is to do a good job in the integration of industry and education. From the analysis of the key words of the selected literature, "integration of industry and education" is the keyword with the highest correlation strength, and at the same time, its frequency is also the highest, so it can be seen that the core position of the integration of industry and education is beyond doubt. From "Anshun college" "university-enterprise cooperation" "training" "cultivation mode" and other keyword analysis, it can be seen that the cultivation of college students is an important purpose of production base construction, in fact many college students have yet to change students mentality, still in the university as a pure students, class, school, and the university, the purpose and to the university to cultivate students. School-enterprise cooperation is the only way for the construction of industry-university-research base. Under this mode, enterprise project experience can be brought into the classroom, giving students a more intuitive feeling, and can stronger strengthen students' technical ability.

How to promote the construction of the integration of industry and education, through school-enterprise cooperation, two-way cooperative education, and the use of the dual platform of

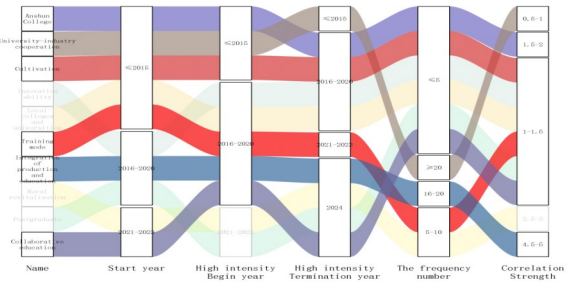


Figure 3 An alluvial map of key words of about the integration of production and education

enterprise and school cooperative education is important. Therefore, the construction of double-teacher(teachers who have been worked in enterprises) teachers and enterprise mentor team is very necessary. In order to better promote the integration of industry and education, bring enterprise mentors into the campus for guidance, or the teaching experience of double-qualified teachers, and integrate new technologies, new processes and new norms into the teaching process, this can achieve the ultimate goal of education base called willing.

Based on this, Sichuan vocational and technical college, launched a double type of teachers, enterprise practice as the guide, supported by enterprise work experience, to create positions as practice, promote the courtyard double teacher team construction, engineering management, for example, 84 in-service teachers, 66 as double teachers, double type teachers reached 78.5%. In addition, the Department of Engineering Management promotes part-time off-campus teachers to participate in education and teaching, with 37 off-campus part-time teachers participating in practical teaching, and 50% of the teachers come from state-owned enterprises. Combined with the development of prefabricated construction and the latest frontier, using prefabricated construction virtual imitation real training base to promote prefabricated and related professional construction, opened a prefabricated construction measurement valuation, prefabricated construction and construction machinery, compiled formed "*Bill of Quantities of Prefabricated Concrete Construction*", "*Project Settlement*", "*Construction Engineering Data Management*" and some compiled materials, combined with the latest practice base and industry provide drawings, optimize the cost, prefabricated professional course theory system, into a line of practice content.

And Southwest jiaotong university hope college based on civil, rail, transportation and other engineering professional the foundation of the construction, southwest jiaotong university hope college according to the teacher experience and enterprise cooperation, promote enterprise visits and experience sharing, participated in the Sichuan city construction consulting group co., LTD., Sichuan housing urban and rural construction exposition, Sichuan is xin agricultural science and technology co., LTD., a series of enterprises to visit and communication activities, lead the teachers to enter the enterprise to learn, get the latest its enterprise development experience. Self-compiled "*Civil engineering materials*", "*Structural Design Principles*" and other course materials, using the latest knowledge to enrich the relevant professional system.

2.2 It must be rooted in local characteristics and strengthen the joint construction and governance of local research and industrial development

In order to implement the industrial construction of IUR base, it is necessary to clear how to better help the industrial development of enterprises and the construction of universities. In fact,

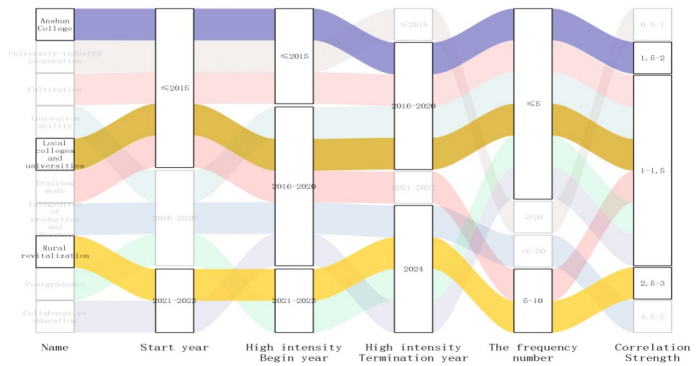


Figure 4 An alluvial map of key words of about the local characteristic

the IUR cooperation requires not only the cooperation of innovation subjects, but also the coordination and integration of innovation goals, organizations, systems and environment[3], the policy support and funding and establishing robust quality assurance systems is important for effectively cultivating skilled talents in vocational undergraduate colleges[4]. These cannot be separated from local resources. Through different modes of cooperation, combined with the guidance of the government and the market demand, this goal can be better achieved. From the perspective of key words, "Anshun College", "local universities" and "rural revitalization" all reflect the achievements of industry, university and research based on local research, as well as the characteristics of close connection between industry, university and research and the construction of local universities.

Using the base construction of IUR and rooted in local resources can effectively promote the optimization of local resources and promote the redistribution of local industrial resources. At the same time, through the localization construction of industry, university and research, it can help the localization of discipline theory construction, promote the enrichment and improvement of theory and the implementation of relevant theoretical experience.

Therefore, a good cooperation mechanism should be established to promote local enterprises and schools to reach cooperation consensus, make use of local enterprise resources and material resources, introduce teaching, design the localization of industrial development path and products based on curriculum design, and the basis of localized development of industry, university and research.

Based on the rail transit development and industry construction requirements of Sichuan, Sichuan construction technology institute colleges deepen cooperation, and Chengdu rydragon machinery manufacturing co., LTD. Outside practice base, promote the drilling blasting method construction of drilling trolley, arch frame support vehicle, concrete wet spray operation vehicle tunnel special operation equipment operation knowledge and skills into the classroom teaching, promote the joint development of drilling blasting method construction "nine car" all intelligent automatic control system. Build an off-campus internship base with China Railway Engineering Service Co., Ltd., and sign the "University-Enterprise Industry-University-Research-Application Strategic Cooperation Agreement" with China Railway Engineering Service Co., Ltd. Strongly promote university-enterprise cooperation and exchange activities, promote China zhongjian design and research institute chief engineer oil xinhua professor senior engineer at the invitation of railway engineering academic forum, and titled "urban comprehensive pipe gallery construction technology innovation and development" of the academic report, to strengthen the Sichuan local rail transit industry enterprise business cooperation.

Southwest Jiaotong University Hope College actively promotes the construction of industry, university and research. Based on the civil transportation industry, it has opened a series of majors such as railway, civil engineering and urban rail, and promotes the construction of industry, university and research in this field. And Wuhu Dawu Trading Co., Ltd., built the virtual simulation training software V1.0, VR security experience software V1.0, HTC VR experience equipment set, with the most advanced virtual technology.

Built the railway overhaul and maintenance virtual training system, with two complete training system, cross-platform to solve teaching pain points, PC & VR end detailed introduces the high-speed railway line maintenance process, through learning, practice, assessment of three modes, let the whole railway virtual simulation maintenance has a thorough theoretical study and training, master the basic theoretical knowledge, safety consciousness and safety maintenance operation. Through these facilities, the application capacity and the demand of local talent construction and development are fully enhanced.



Figure 5 VR security experience training system and training site map

2.3 Must focus on rural revitalization and study the policy mechanisms of urban-rural coordination and rural issues

With the gradual increase of urbanization rate and the stabilization of urban population, the hollowing out of rural areas has become an irreversible trend. However, based on the influence of rural tourism, rural pension and food security, rural development is not only an important focus of urban and rural development, but also a part that cannot be ignored in the future development. Rural revitalization is also becoming much attention in the research of industry-university-research. In the construction of industry-university-research base, rural revitalization has high attention intensity and continuous attention, and has become a cluster alone. In the key words, agriculture is also a key word is unavoidable.

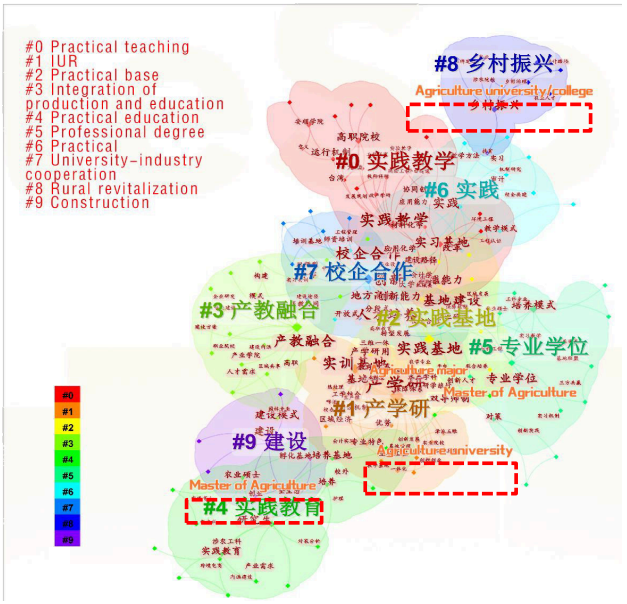


Figure 6 Keyword group classification diagram

In fact, there is no lack of policy support for rural construction, but the lack of construction talents, limited support funds, difficult policies, imperfect modernization and other problems, have been restricting the development of rural construction. The solution of rural problems needs a process, especially at the present stage of China has shifted from high-speed development to high-quality development stage, each construction fund needs to be used on the edge. At the same time, the relocation of rural population restricts the development of rural infrastructure and rural

industries. The newly born "second generation of farmers" and "third generation of farmers" are reluctant to farm, and rural land is abandoned.

Therefore, the development direction of the construction of industry-university and research base focuses on the rural revitalization and the policy mechanism of urban and rural revitalization, and the study of industry-university and rural coordination and rural issues. Majors related to agriculture, forestry, construction, marketing and administration can be scientifically combined with the development needs of rural revitalization, landed in rural areas, promote rural revitalization and urban-rural integration, promote scientific and technological innovation from "IUR" to "government-industry-university-research(GIUR) and application", and promote the construction of IUR bases.

In this respect, in fact, many agricultural colleges, as the key words suggest, are actively exploring. As the main position of agricultural talent training, agricultural colleges and universities have unique advantages in helping the revitalization of rural talents[5]. In the process of rural model house design, and the process of rural infrastructure construction, it can be applied, especially some equipment majors can study rural revitalization and agricultural modernization, and study the use of new equipment in rural areas. However, due to the hollowing out of rural areas, aging, as well as rural funding problems, these ways are very difficult to promote. In this regard, Sichuan Institute of Architecture and Technology has promoted the summer social practice activity of "Promoting rural Revitalization", and the social practice activity of "Three to the countryside" in the summer vacation of 2021-2024 has also been carried out smoothly. Southwest jiaotong university hope college is based on the rural construction, in 2020, the town of rural revitalization project training, 2022 continuous support to the town, accurate poverty alleviation work by media reports and by the Sichuan provincial education committee, Sichuan provincial education department as "excellent collective", college of poverty alleviation cadre Xia Fuqiang comrade won the reward.

2.4 Must devote ourselves to scientific and technological innovation and explore the training mode of higher talents and professional talents

The cooperation mode of IUR institute itself is a mode to promote industrial upgrading by skillfully using the scientific research level of the university and the development of social science and technology. The construction of IUR should think of the effect, and it is not enough to bring students to the base for internship. Simple ones, such as engineering students draw a few pictures and agricultural students do a few experiments, can not produce good results. At the same time, it can not mobilize students' thinking, nor let the IUR and research play its due effect. At present, although there are various ways of students' course assessment, and students' education also has high practice hours, it does not mean that these practice hours and many practical assessment methods can help students to achieve the ability to complete tasks independently in the future work.

Therefore, in order to give full play to the effectiveness of IUR mode, to allocate scientific and technological talents who can conform to the industrial construction, and at the same time, to cultivate a wave of students with practical ability, scientific and technological innovation must be committed to. Innovation is the primary productive force and the driving force of progress. The emergence of keywords such as "master", "practical teaching", "training mode" and "practical base" means that the separation between industry, education, research and innovation is increasingly prominent. The emergence of "master" in recent years thinks that this highly intelligent innovative thinking and ensuring the landing of graduate ability are the direction of talent training of industry, university and research in the future. It is true that there are certain differences in the function and direction of talent training of vocational education and that of higher education, but the rapid development of industry under the background of industry 4.0 also requires the rapid integration of

professional talents into the society. Both advanced talents and professional talents are inseparable from the cultivation of innovation ability.

Therefore, it is necessary to consider the reasonable methods of integrating the curriculum of higher talents and professional talents into IUR, combine the education, teaching and assessment of the curriculum with the scientific thinking of industry-university and research, and explore a talent training mode in line with the construction of IUR base.

Sichuan Vocational and Technical College of Architecture promotes the friendly school-enterprise cooperation mode. By tripartite agreement, teachers in the industry will give part-time teachers as part-time teachers. Through communication with friendly units, a perfect directory of teachers has been established, which can be updated and maintained for enterprise teachers every year. Encouraging corporate tutors and campus tutors to guide graduates and optimize the quality of graduates. This mode is simple to operate, convenient to move, convenient for teachers to cooperate directly, and can quickly replace inappropriate teacher resources. It is a relatively simple way to operate. In the school, we have completed the construction of BIM Basic Technology Center, prefabricated building virtual imitation training base, green Building Evaluation and Research Center, and comprehensive Training center for Engineering management. With the school base and training center, in-service double-teacher and part-time teachers are used to bring students to complete the practical training teaching. Outside the school, the use of the listed practice base, bring students into the base for teaching,

Southwest Jiaotong University Hope College adopts the model of learning and competition.50% of the old teachers in the college have working experience in enterprises, and many of them have the background of enterprise cooperation. In promoting the construction of school-enterprise cooperation base, the enterprise project is mainly driven by the teachers of the college, and relies on the enterprise to become the project base. At the same time, relying on the BIM training center on campus, the BIM studio serves as a fixed working place to promote project research. In the project, teachers are the main ones and students are assisted, so that student teachers can directly have a platform to participate in the project and form the results of the consulting project. After forming the results, students are actively encouraged to participate in the field competition, through the preliminary results and simulate the project process, to achieve the secondary consolidation of knowledge and software expansion, to ensure the depth of the integration of education and teaching and practical work.

3 Summary of the mode and conditions of school-enterprise joint training base construction

3.1 Characteristics and positioning of the training base under the background of industry and education integration

Based on the theoretical and practical results, in order to build a good IUR base, we must implement the characteristics of the IUR base, so as to well carry out the construction of the IUR base. From the above analysis, it can be seen that the construction of the IUR base must be based on the following characteristics:

First, it is practical and practical. The training base takes the actual working environment and process as the template to provide students with training close to the practical work and emphasize the practical application of skills. Second, resource sharing. The training base jointly built by schools and enterprises should make full use of the educational resources of schools and the work resources of enterprises, and when necessary, cooperate with the government to realize complementary advantages. Third, dynamic update. The content and method of practical training can be

dynamically adjusted according to the development of the industry and the needs of enterprises to ensure the synchronization of education and the industry. The fourth is double-teacher teaching. The teaching in the training base is usually jointly undertaken by school teachers and enterprise experts, forming a "double-teacher" teaching mode. Fifth, the integration of achievements, the training base has become a platform for the combination of industry, university and research, promoting the transformation of scientific research achievements, and providing technological innovation support for enterprises.

And the training base under the industry, university and research positioning. First, the talent training base, through the training base is an important place to cultivate high-quality technical talents, emphasizing the combination of theory and practice, can produce directors of the talent. The second is the industrial docking window, as a window for the docking between schools and enterprises and industries, which helps the school to understand the industrial dynamics and adjust the professional Settings and teaching content. The third is the technology innovation platform. The training base can become a platform for both the university and the enterprise to carry out technological innovation and product research and development, helping the school to promote the transformation and application of technological achievements. The fourth is the social service platform, the training base can also provide technical training, vocational qualification certification and other services for the society, to enhance the school's ability to serve the society. Fifth, the corporate culture experience area. Through the practical training, students can experience the corporate culture in advance, which is conducive to the development of their professional quality and the adaptation of their future employment.

3.2 The needs and interests of school-enterprise cooperation

School-enterprise cooperation must be a mutually beneficial relationship, both sides gain something to meet their own needs, and obtain the desired resources and benefits.

From the perspective of the university, talent is the foundation and key, and the training of innovative talents in industry-university-research cooperation plays a core position and plays a key role in the process of linking innovation linkage and building an innovative country in China[6], cultivating talents is the first purpose of school-enterprise cooperation in universities. Cultivating talents who can have employment ability and improving students' employment rate and employment quality, which is the demand of the university. Therefore, starting from the fundamental requirements and promoting cooperation is required. But light is not enough, campus teachers due to long-term and social disconnection, can cause backward education knowledge, education concept old, through university-enterprise cooperation can also improve the quality of education, through the enterprise actual case and project, strengthen teaching practical and practical, and enhance teachers' practical ability, promote the "double type" teacher team construction. Finally, the cooperation between schools and enterprises can obtain enterprise resources, such as capital, equipment, technology, etc., so as to improve the teaching and research conditions and give the school some substantial benefits. Through such cooperation, schools can get a lot of revenue. The first is to enhance the school's brand image and social recognition, to attract more high-quality students. Secondly, by promoting the transformation of scientific research achievements, enhance the scientific research ability and economic benefits. The third is to establish a stable off-campus practice base to provide students with practice opportunities. The fourth is to increase the connection with businesses to bring potential donations and funding to schools.

From the perspective of an enterprise, the first priority is to obtain technical and skilled talents in line with the needs of the enterprise. Social recruitment has certain instability. At the same time, there are certain risks in cultivating new talents, and there are many job-hopping talents after they

are mature. School-enterprise cooperation can reduce the cost of recruitment and training, and cultivate stable potential talents. Secondly, use the school's research resources and innovation ability to carry out product research and development and technological innovation, constantly bring forth the new, or strengthen the business scope of the enterprise. Third, through school-enterprise cooperation, it can help enterprises to open up their popularity, obtain better communication in schools, improve the social image of enterprises, and enhance the brand competitiveness.

Finally, through school-enterprise cooperation, enterprises can also understand the industry development trend and cutting-edge technologies, and improve the competitiveness of enterprises. And in this process, the enterprise can also obtain certain benefits. First, it can give priority to the selection of outstanding graduates and reserve human resources for the development of enterprises. Second, through participating in the education process, customized training of talents in line with the specific needs of enterprises, to meet the production needs of enterprises. Third, use the university's scientific research facilities and expert team to reduce production costs.

In general, in terms of interests, school-enterprise cooperation should match the supply and demand of talents. School-enterprise cooperation can achieve the matching of the supply and demand of talents. There schools and enterprises can achieve mutual benefit and win-win results through resource sharing. Schools can obtain funds, equipment and technical support from enterprises, while enterprises can use the scientific research and human resources. The third is common development. School-enterprise cooperation helps both sides to grow together. Schools can adjust the teaching content through enterprise feedback, while enterprises can improve their innovation ability through school-enterprise cooperation. The fourth is risk sharing. In the school-enterprise cooperation, both sides jointly bear certain risks, such as talent training.

3.3 Functions and requirements of the training base for schools and enterprises

Starting from the needs and interest relationship of school-enterprise cooperation, it is easy to get the functions and needs of the training bases and enterprises.

Stable and mature talent training mechanism. It is necessary to establish a stable cooperation mechanism, train talents, ensure the full utilization and stability of education and teaching, and at the same time serve enterprises, so as to make the long-term effect of education and teaching. Therefore, it is necessary to have a stable system and mechanism as the basis for all-round research.

The function is that there must be a site and space to meet the needs of the training base. The space is a space carrier to support students and teachers 'teaching, an important place to meet the needs of practical teaching, scientific research and technology development, and plays an important role in cultivating students' ability of connecting theory with practice, innovative spirit and application ability.

Modern industrial production facilities for education and teaching 1 production. The demand of industry-university-research education and teaching is clearly pointed out that it is to be able to cultivate students who can be connected with their posts, so the practice base posts must have production facilities or similar project conditions that can be connected with the practice industry, and give students education, teaching and learning.

Enough matching enterprise teacher or double-qualified guidance teacher teaching. In order to achieve such a talent training mode, the school needs to continuously deepen and improve the education level of teachers, ensure that the teaching content is in line with the actual needs of enterprises, ensure the improvement of students' comprehensive quality, so that students can get all-round development in theoretical knowledge and practical skills.

3.4 Factors affecting school-enterprise cooperation

It should be said that different schools have cooperation modes of different schools. As the aforementioned analysis shows, school-enterprise cooperation must be rooted in local characteristics and analyzed according to the situation of different schools. The research team comes from different teachers from different universities, and analyzes and discusses according to the relevant experience summary.

The first is the impact of the policy environment. At present, the state encourages the construction of double-teacher team, and various schools are also encouraging the construction of double-teacher teacher team. Taking Sichuan Vocational and Technical College of Architecture as an example, through the tripartite cooperation agreement, teachers are encouraged to enter the enterprise post practice. Double-qualified teachers are divided according to the enterprise experience. According to the work experience of teachers and the work level of the enterprise, double-qualified teachers are classified and included into the professional title assessment and academic administration assessment, so as to improve the management.

At the same time, double-qualified teachers and part-time teachers in enterprises have also become the requirements of assessing the special establishment or the undergraduate level construction of the school, and become the heavy driving force to promote the quality of education.

The second is the cooperation mechanism with off-campus enterprises. A cooperation mechanism is very important, and having a stable partner is beneficial. Sichuan Institute of Architecture Vocational and Technology has actively established its own part-time teacher team. According to the policy requirements, it selects part-time teachers from the aspects of educational level, working experience, professional qualifications and industries, and establishes a teacher directory. Teachers are invited to conduct in-post practical teaching and participate in theoretical knowledge teaching, integrate theory and practice, enrich the course teaching mode, bring practical experience into the classroom, and give remuneration, so as to expand their thinking and ensure operational interests.

The third is the level of economic support. Third, the level of economic support. At present, the economic situation is in the transition stage, and the support from enterprises and the government will directly promote the development of the training base and ensure the sustainability of school-enterprise cooperation. However, in addition to the encouraging policies, the relevant economic support is limited, which limits the scale and depth of the training base to some extent. Basically, it will be more common for part-time teachers to teach local cooperation, because it can effectively control costs and improve the efficiency of cooperation. However, this also means that it is difficult to find relevant teachers and enterprise resources for remote tutoring. To find a better platform and information, it is undoubtedly a huge challenge for those schools with relatively remote geographical locations and relatively scarce resources.

The fourth is the strength of schools and enterprises. The strength level of schools and enterprises directly affects the construction and operation effect of the training base. Strong schools can provide better teaching resources and scientific research platform, while strong enterprises can provide more advanced practical training equipment and technical support. This mode of strong combination can effectively improve the overall level of the training base and cultivate high-quality talents more in line with the market demand.

However, for those relatively weak schools and enterprises, cooperation may face more challenges, such as insufficient resources, backward technology and other problems. Therefore, to enhance their own strength and strengthen the connotation construction has become the common goal of schools and enterprises, so as to ensure the sustainable and healthy development of the training base.

4 Future work

4.1 Summary of the current situation research experience

In general, it seems preliminary that based on the analysis of the two schools of the research team and the relevant schools in Sichuan Province, it can be seen that China's industry, university and research practice has a relatively perfect experience, but the system is still continuing to explore and improve.

It can be said that in recent years, China has been vigorously promoting the work of education and teaching, and has cultivated a number of high-tech enterprises with strong practical ability and the latest industrial technology through industry, education and research. At the same time, the national government has also given certain subsidies to high-tech industries, and given sufficient policies for the development of industry, university and research. Starting from different colleges and universities, due to different colleges and universities have different positioning strength, different economic foundation and different resources, the progress of the school-enterprise joint construction practice base is different, and the implementation mode is also different. At present, the track of industry, university and research in rural revitalization still needs to be improved, but other aspects have been explored. In addition, the university-enterprise cooperation base construction in part of colleges and universities, although improve the infrastructure, but in terms of scientific and technological achievements transformation remains to be improved, although the domestic production base construction of employment training results to the naked eye objective, but employment form remains to be improved, especially the employment of private schools also need more comprehensive further promote mode optimization.

Behind this is the need to explore and improve the optimization of the industry-university-research system and the model of university-enterprise base construction, which requires a good set of evaluation mechanism and system suggestions. There is a good industry-university-research evaluation mechanism to evaluate the benefits of all aspects of the school-enterprise cooperation base, so as to optimize the effect of school-enterprise co-construction base construction, continuously improve the service capacity and efficiency of industry-university and research institutes, promote the transformation of scientific research results, and achieve the desired effect.

4.2 Directions and recommendations for further research

The school-enterprise joint training base is an important form of the integration of industry and education, which has a positive impact on the economic benefits, educational benefits and social benefits. If there is no relevant benefits, the construction of industry-university and research is meaningless. Industry-university and research is a systematic complex work, involving multiple subjects and is a composite system work. The practice base jointly built by universities and enterprises should have certain benefits, and the construction of industry, university and research, and the systematic evaluation of industry, university and research, can continuously improve and help improve the efficiency of industry, university and research.

First is the economic benefit, can be seen from the past experience, economic conditions and level is an important problem restricting the development of enterprises, from the economic perspective of analysis, reduce the cost of enterprise talent training, improve enterprise technology innovation ability, improve product quality, enhance enterprise market competitiveness, promote enterprise brand promotion, these are the main economic benefit, enterprises can through training base and school cooperation to cultivate talents, avoid training the time and cost. But, how to measure it and achieve an effective assessment? The most intuitive economic benefit is reflected in the income. For enterprises, it is the income of the participants, the income of part-time teachers,

and the proportion of production output income. For the school, it is the economic input, the output value of scientific research results, which are some intuitive economic indicators.

The second is educational benefits. The training base provides students with a real production environment, helps students to apply theoretical knowledge to practice, and improves their practical ability. The effect should be specific benefits. The training base can also help students understand the corporate culture, work process, etc., and cultivate students' professional quality and teamwork ability. The training base can combine the teaching content of the school with the actual needs of enterprises to improve the quality of talent training and meet the needs of enterprises. Teachers can participate in the teaching and scientific research activities in the training base to improve their practical teaching ability and better serve students. Therefore, the specific indicators should be selected firmly around the above elements of talent and teacher training. Proportion of curriculum practice integration, awards of students or teachers in competitions, proportion of double-teacher talents, effect of curriculum construction, curriculum system construction, operation of 1 + X certificate operation, teaching quality satisfaction, scientific research achievements, etc.

Finally, social benefits. The joint training base can cultivate more talents in line with the needs of enterprises, promote employment, and promote the specific improvement degree of employment, and produce results. In addition, the training base can promote the transformation of scientific and technological achievements, the contribution to the local economy, the promotion of local economic development, the promotion of industrial reform, social influence, and the dissemination of culture are all specific social benefit factors.

Funding

"Key Research Base of Philosophy and Social Sciences in Sichuan Province- -Sichuan Education Development Research Center of China West Normal University"--Take the specialized course of Principles of Structural Design as an example (CJF23070); China Society of Electronics and Labor "industry and education integration, school-enterprise cooperation" education reform and development topic, NO.Cea12023016. Project of Sichuan Private Education Association (Research Center) —— Research on "digital empowerment" teaching mode and evaluation reform in private university classroom- -Take the course of "Principles of Structural Design" as an example (MBXH23YB440)

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References

[1] Tao Hu, Shen Li. The enlightenment of foreign innovation and entrepreneurship education mode to Chinese universities [J]. Chinese university teaching,2013,(02):91-94+90.

[2] Fu Junchao. Operation mechanism and performance evaluation of industry-university-research cooperation [D]. China University of Geosciences, 2013.

[3] Rao Yanting. The connotation, requirements and policy conception of "industry-university-research" collaborative innovation [J]. Higher education exploration, 2012,(04):29-32.

[4] Kan S , Pek S L , Linli Y . Research on the Path of Cultivating High-Level Technical and Skilled Talents in Vocational Undergraduate Colleges and Universities[J].Journal of Contemporary Educational Research,2024,8(7):150-157.

[5] Hui Zhidan. Research on Serving Rural Talents in Agricultural Universities under the background of rural Revitalization strategy [D]. And Huazhong Agricultural University, 2020.DOI:10.27158/d.cnki.ghznu. 2020.000483.

[6] Chu Guogang. Research on innovative talent training mode and mechanism of industry-university-research cooperation [D]. Harbin Engineering University,2018.